

Modelling Football Match Events and Player Connections: Bayer Leverkusen 2023/24

Project Codename: Beyond the Scoresheet

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ABSTRACT

This proposal tries to solve a data-modeling problem using a real, public sports dataset: showing how matches, teams, players, lineups, and timestamped match events are linked across all 34 Bundesliga games that Bayer 04 Leverkusen played during its historic, undefeated 2023–24 title-winning season.[1]. Using StatsBomb’s Open Data (competition ID 9, season ID 281) [2, 3], we study how relational and semantic (RDF/OWL) representations preserve and expose the same relationships — such as a goal’s link to its scorer, lineup context, and an explicitly source-coded assist pass — for a shared set of relationship-oriented queries. The project’s contribution is a way to show and rate things, not a way to predict or explain why the club won.

KEYWORDS

Sports Event Data Model, Relationship Modelling, Semantic Interoperability

1 INTRODUCTION

A football game involves many interacting factors, such as who participated, where they played, and what they did, as well as shots, goals, cards, replacements, and how everything worked together to decide the result. This study examines the 2023–2024 Bayer Leverkusen Bundesliga season. Key domain entities include matches, teams, players, positions, formations, and discrete match events [4]. The relationships between entities make this domain difficult to represent: a goal cannot be separated from its scorer, assist, passing sequence, pitch formation, or substitutions [5]. This project addresses the data-modelling problem of how to represent the events of a football match — the players and teams involved, their actions, and the relationships that connect events across a match or a season — in a way that is precise, relationally complete, and reusable across information needs [6], so that humans and computational systems can reliably reconstruct who was involved in any given event, trace how one event led to another [7], and compare events.

2 MOTIVATION

An integrated representation of football event data can link player, match, lineup, and event information that is currently scattered among separate records. This provides a unified view of player participation, positional duties, substitutions, and subsequent actions across games, and enables more complex queries that trace player–event interactions while keeping their time and match context. Football analysts and scholars may find this useful for quicker

retrieval, comparison, and reuse of detailed match information. Explicitly expressed relations also provide a cleaner base for automated querying and knowledge extraction by computational systems. Hence, this study explores ways to represent related football information while retaining key contextual linkages.

2.1 Motivating Example

Let’s say a Leverkusen player comes on as a replacement and creates a chance to score. To understand what that player did, several records must be linked: the substitution, the player’s position in the game, the next pass or shot, and the game in which it happened. Current event records capture these facts in isolation, but the relationships between them are not explicitly modelled as one connected journey, which makes it difficult to track a player’s path from joining the match to creating an opportunity. This project encodes these linkages so that such connected occurrences can be queried jointly, while keeping their temporal and match-specific context.

3 TEAM JUSTIFICATION

Our team is composed of three people with complementary knowledge. All participants have shared coursework experience in database architecture and data administration. One member has football domain expertise, another has hands-on experience with semantic web technologies, and another has hands-on experience with applied artificial intelligence. This combination supports domain analysis, data modelling, semantic representation, execution, and evaluation of the project. Every member will be involved in all project phases, and progress will be documented via our shared GitHub repository and commit history.

REFERENCES

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